

Static vs Contextual Embeddings for Lexical Ambiguity

A Word-in-Context Evaluation

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Introduction

Many words have more than one meaning. A **bank** can be the side of a river or a place that keeps money. This is **lexical ambiguity**. Computers represent words as vectors, so we ask whether those vectors distinguish meanings in context.

Background

Homonymy means unrelated meanings, as in river bank and money bank. **Polysemy** means related meanings, as in paper as material and paper as a newspaper. A **word embedding** is a vector learned from language. **Cosine similarity** measures how closely two vector directions align. **Static GloVe** gives each word one vector in every sentence [2]. **Contextual BERT** creates a new vector for each occurrence using its sentence [3].

Related work

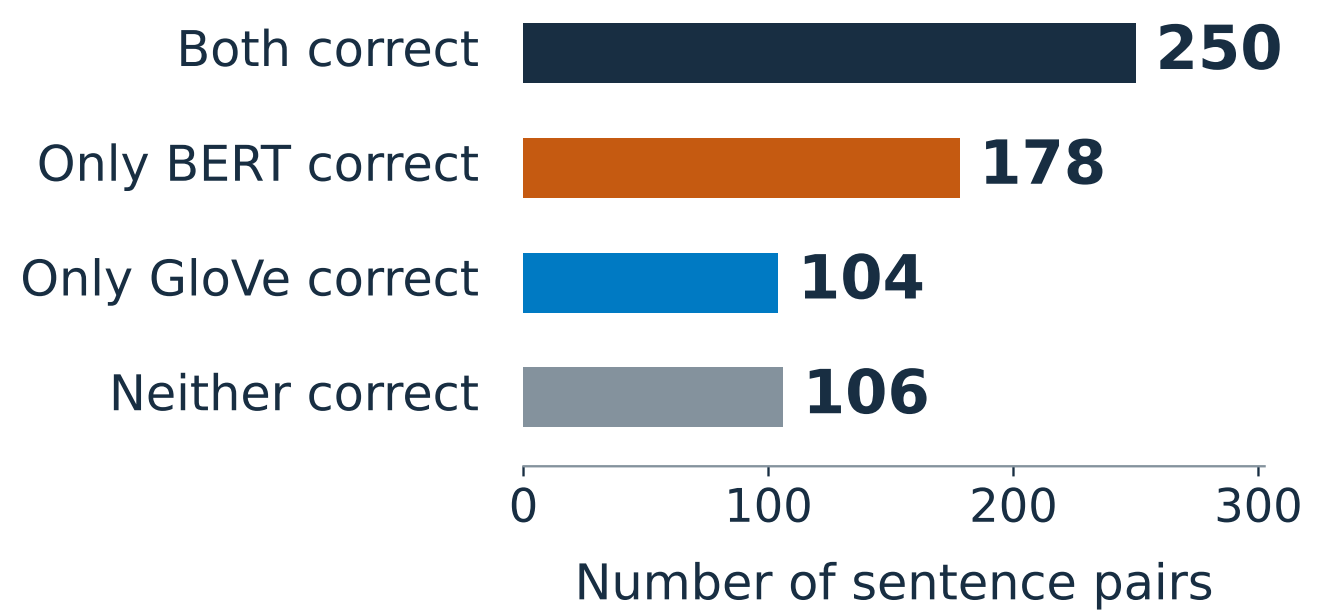
Earlier word sense disambiguation used dictionaries such as WordNet. Multi-sense embeddings assign several vectors to a word [4]. ELMo and BERT instead build representations from context. WiC tests whether the same word expresses the same meaning in two sentences [1]. Most competitive systems are fine-tuned. We study frozen representations.

Hypotheses

H1. A static target vector cannot distinguish senses because both uses receive the same vector.
H2. Averaging nearby GloVe words should add a small contextual signal.
H3. Frozen BERT should outperform that GloVe context baseline because its target vector changes with the sentence.

Methodology

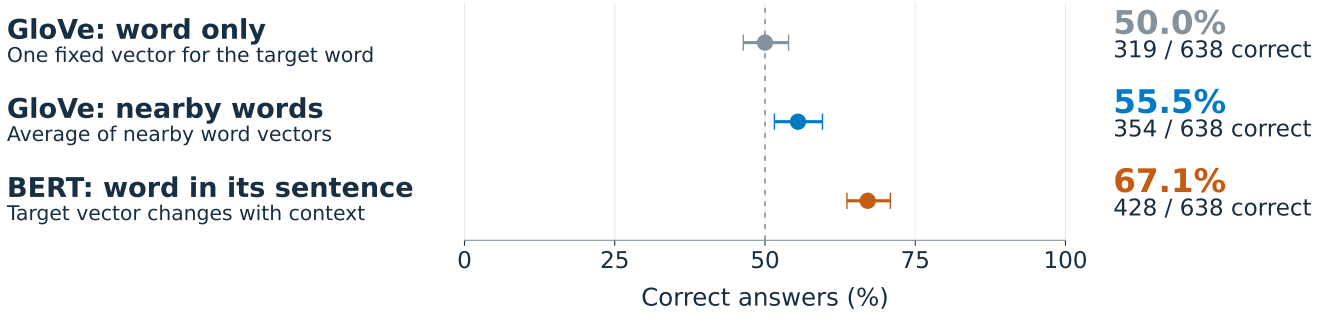
Data. SuperGLUE WiC contains 5,428 training pairs, 638 validation pairs and 1,400 public test pairs without public labels.
Systems. We compare the GloVe target word, the average of nearby GloVe vectors within ± 2 , and bert-base-uncased target vectors averaged over the last four layers.
Decision. Each sentence pair receives a cosine similarity. A training-selected threshold turns it into “same meaning” or “different meaning.” Window size, layers and thresholds use training data only. Validation is evaluated once.
Uncertainty. Paired confidence intervals use 1,000 bootstrap resamples.



Paired outcomes show that the systems make different mistakes.

Results

System	Acc.	F1	AUC
GloVe target	50.0%	33.3%	0.500
GloVe context	55.5%	55.1%	0.580
BERT	67.1%	66.7%	0.716



BERT is **11.6 points** more accurate than GloVe context. The paired 95% interval is 6.6 to 16.9 points. Layer 9 reaches 66.6%, while layer 12 reaches 63.6%.

Discussion

BERT alone gets 178 pairs right. GloVe context alone gets 104 right. BERT recovers 77.4% of same-sense pairs but only 56.7% of different-sense pairs. GloVe context shows the reverse tendency: 46.4% and 64.6%. Both systems fail on difficult cases such as “engrave a letter” versus “engrave a pen.”

Conclusion

Contextual embeddings capture sense differences better than static representations on this WiC protocol, even without fine-tuning. Yet BERT still makes 210 errors. The finding is limited to one frozen model, a simple GloVe baseline and 638 validation pairs.

References

[1] Pilehvar & Camacho-Collados (2019). *WiC*. NAACL.
[2] Pennington, Socher & Manning (2014). *GloVe*. EMNLP.
[3] Devlin et al. (2019). *BERT*. NAACL.
[4] Camacho-Collados & Pilehvar (2018). *From Word to Sense Embeddings*. JAIR.
[5] Loureiro et al. (2021). *Language Models for Word Sense Disambiguation*. Computational Linguistics.
[6] Haber & Poesio (2021). *Polysemy and Homonymy in Contextualised Language Models*. Findings of EMNLP.